

HONOURED

HEALTHKIT ACCURACY — WATCH VS PHONE

Feature Update — Filed Sept 2026

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Core issue

HealthKit aggregates step data from multiple sources — iPhone motion sensor, Apple Watch, and any connected third party device. This can result in significant overcounting. A user's iPhone in their pocket records every incidental movement as steps, while a dedicated wrist-worn device like a Garmin or Apple Watch applies sports-grade algorithms tuned specifically for step accuracy. For Honoured — where data is proof — the source of that data matters.

1. The Problem

During real-world testing, a significant discrepancy was found between Garmin watch step count (7,800 steps) and HealthKit step count (10,900 steps) for the same period — a difference of approximately 3,000 steps. This is not a rounding difference. It is a meaningful gap that could allow a user to receive an HONOURED stamp on a commitment they did not genuinely meet.

This directly undermines Honoured's core philosophy: your data is proof, automatic and unquestionable. If the data source is inaccurate, the proof is inaccurate.

2. Why The Gap Exists

- iPhone in pocket — picks up every movement. Reaching, shifting, incidental motion all register as steps. The phone cannot distinguish purposeful walking from background movement.
- Garmin watch — purpose built for movement tracking. Sits on the wrist, measures arm swing and stride with a dedicated sports-grade accelerometer. Filters incidental movement.
- Apple Watch — significantly more accurate than iPhone, closer to Garmin. Wrist-based detection filters pocket noise.
- HealthKit aggregation — by default HealthKit combines data from all available sources. If both iPhone and Apple Watch are present, it attempts to deduplicate but does not always succeed.
- Result — users with phone in pocket and no Apple Watch may see significantly inflated step counts in Honoured.

3. Accuracy Comparison

Source	Accuracy	Why
Garmin Watch	Highest	Purpose-built sports sensor, wrist-based, proprietary algorithm refined over decades
Apple Watch	Very High	Wrist-based, dedicated accelerometer, very close to Garmin accuracy
Android Watch / Wear OS	Very High	Wrist-based, comparable to Apple Watch, filters incidental movement
iPhone (pocket)	Lower	Picks up all movement, cannot filter incidental motion reliably

Android Phone (pocket)	Lower	Same limitations as iPhone — overcounts incidental movement
HealthKit aggregate	Variable	Depends on which sources are connected and active
Google Fit aggregate	Variable	Same aggregation issue as HealthKit — source priority matters

4. The Fix

Two levels of improvement — near term and future roadmap:

Near term — HealthKit source priority:

- HealthKit allows specifying a preferred data source. Vu to set Apple Watch as the priority source for step data over iPhone motion sensor.
- When Apple Watch is present — use Apple Watch data only. Ignore iPhone pocket counting.
- When no Apple Watch is present — use iPhone as fallback with a note to the user that accuracy may vary.
- This is a relatively small code change with significant accuracy improvement for Apple Watch users.

Future roadmap — Garmin Connect API:

- For users on Garmin watches — integrate Garmin Connect API as a direct data source.
- Garmin data bypasses HealthKit aggregation entirely, pulling the most accurate step and distance data directly.
- For Android users — integrate Google Fit API as the equivalent of HealthKit. Set Android Watch / Wear OS as preferred source over phone.
- Android Watch accuracy is comparable to Apple Watch — wrist-based, purpose built, reliable.
- Android Phone in pocket has the same overcounting limitations as iPhone — avoid as primary source.
- This is the watch integration roadmap item already filed — Q2 2027 Apple Watch, Q3 2027 Garmin and Android.
- Until then, Apple Watch as preferred HealthKit source is the best available accuracy for V1.x.

5. Why This Matters for Honoured

Honoured's entire value proposition is built on unquestionable proof. 'Your Apple Health data doesn't lie' only holds if the data source itself is trustworthy. A user who commits to 10,000 steps and receives an HONOURED stamp based on inflated iPhone pocket counting has not truly been held accountable. They have been given false proof.

Fixing the data source priority is not a cosmetic improvement — it is a philosophical one. It is the difference between Honoured being a genuine accountability tool and a well-designed app that tells people what they want to hear.

6. Developer Note for Vu

- Set HKBiologicalSexCharacteristics preferred data source for step count to Apple Watch when available.
- Use HKSourceQuery to identify available sources and prioritise wrist-based devices.
- Display data source label in the AT A GLANCE tile — 'Apple Watch' or 'iPhone' — so users know what they're seeing.
- Consider a user-facing accuracy notice in Settings: 'For best accuracy, use an Apple Watch or connect a compatible wrist-worn device.'

7. User-Controlled Data Source Selection

Garmin and Apple Watch are well documented as very close in accuracy for step counting and everyday distance tracking. Neither is definitively superior for Honoured's core use case — the difference is subjective and

device-dependent. A serious Garmin athlete knows their watch is most accurate for them. An Apple Watch user trusts their device. Forcing a default onto users who know their own setup better than the app does is the wrong approach.

The solution: let the user choose their preferred data source in Settings. This respects their knowledge, improves trust in the data, and future-proofs the watch integration roadmap — the selector exists in Settings before Garmin and Android Watch are fully connected.

Proposed Settings UI — Data Source Preference:

Option	When to use
Let Honoured decide (default)	App selects highest accuracy available — Apple Watch over iPhone, Android Watch over Android
Apple Watch	User has Apple Watch and wants wrist-based data only
Garmin	User has Garmin watch connected — uses Garmin Connect API (future roadmap)
Android Watch / Wear OS	User has Android watch — wrist-based, high accuracy (future roadmap)
iPhone	No watch available — user understands accuracy may vary
Android Phone	No watch available — user understands accuracy may vary

Why this matters for Conquered:

In Conquered, a portion of the phone is disabled as a consequence when commitments are broken. If step or distance data is inflated by phone pocket counting, a user could satisfy a fitness commitment on paper while Conquered enforces a consequence based on numbers that aren't real. User-controlled data source selection ensures the integrity of both the commitment and the consequence. Accurate data is not just a feature — it protects the philosophy.

Filed Sept 2026 based on real-world testing. Priority: medium — address in V1.2 polish sprint. Near term fix: set Apple Watch as preferred HealthKit source for step and distance data. Android and Garmin integration deferred to watch roadmap (Q2-Q3 2027). Accuracy hierarchy: Garmin ≈ Apple Watch ≈ Android Watch >> iPhone/Android phone in pocket.